

# Mayank Bansal

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## EDUCATION

**Lovely Professional University** | B. Tech in Computer Science & Engineering | *Punjab, India*

*Expected Graduation: May 2028*

- **TGPA:** 9.33 / 10.0 (2nd Year)
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Probability & Statistics, Database Systems

## TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, HTML5, CSS3
- **Machine Learning:** TensorFlow, Scikit-learn, NumPy, Pandas, Librosa, TF-IDF, Matplotlib
- **Web Development:** Flask, REST APIs, Gunicorn, Render Deployment, UI Design
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook

## FEATURED PROJECTS

**WasteSnap – Real-Time Waste Classification** | *MobileNetV2, Flask, JavaScript, Render*

- Developed and deployed an AI-powered application to classify 9 types of waste using a fine-tuned **MobileNetV2** architecture, achieving ~61% validation accuracy on a custom dataset.
- Engineered a production-ready **Flask** backend with **Gunicorn** on **Render** to handle image uploads and live camera feeds.
- Implemented real-time recyclability tagging and disposal tips on the frontend to drive eco-friendly user behaviour.

**DigiDetect – Handwritten Digit Recognition** | *TensorFlow, Flask, CNN*

- Built a full-stack web app featuring a custom **Convolutional Neural Network (CNN)** that achieved **98.9% test accuracy** on the MNIST dataset.
- Designed a dynamic canvas interface for users to draw digits, processing inputs via a Flask API to return real-time predictions and confidence scores.
- Implemented robust image preprocessing pipelines to ensure consistent model performance on user-generated inputs.

**Music Genre Classification** | *Librosa, Flask, Audio Processing*

- Architected a backend API capable of accepting MP3/WAV files and extracting complex audio features (MFCCs, spectrograms) using **Librosa**.
- Solved deployment challenges related to **CORS** and **NumPy** version compatibility to ensure scalable, error-free inference.
- Integrated the model with a responsive frontend to visualize genre predictions for uploaded audio tracks.

**Mood2Mail – Real-Time Email Tone Analyzer** | *TF-IDF, Naive Bayes, Flask*

- Created a real-time tone detection system utilizing **TF-IDF** vectorization and **Naive Bayes** to classify email drafts into 5 distinct tone categories.
- Developed a seamless user experience using **JavaScript** fetch APIs to provide live tone feedback as the user types.

## CERTIFICATIONS

- **Complete Data Science and Machine Learning Skill Up** – GeeksForGeeks
- **AI Tools Skill Up** – GeeksForGeeks
- **Elements of AI** – University of Helsinki